

Machine Learning-Based Crime Category Classification and Spatio-Temporal Pattern Analysis

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Abstract—Crime records increasingly guide policing resources, yet many published classifiers report accuracy alone and say little about where or why a model fails. This paper classifies 1,060,801 Chicago crime incidents (2020–2024, City of Chicago Open Data Portal) into four categories – Violent, Property, Drug/Public Order, and White-Collar – from eleven district-, location-, and time-based features, with no victim demographic inputs. Five classical classifiers and three neural baselines (a multilayer perceptron, an FT-Transformer, and a graph convolutional network over district adjacency) are trained under matched unweighted and sample-weighted configurations and scored with accuracy, macro and weighted F1, MCC, and one-vs-rest ROC-AUC. XGBoost performs best in the unweighted configuration (67.5% accuracy, macro-F1 0.51). Paired McNemar tests and bootstrap confidence intervals show that it significantly outperforms the Decision Tree, Logistic Regression, Naive Bayes, FT-Transformer, and graph network ($p \leq 0.006$), while the MLP performs comparably in the unweighted configuration ($p = 0.47$); Random Forest was not included in the paired tests. The two rarest classes obtain the highest AUCs (0.88 White-Collar, 0.86 Drug/Public Order) despite low F1, showing that their weakness is a thresholding and base-rate effect rather than missing signal. The paper contributes a reproducible, demographic-free pipeline that separates ranking quality from decision-threshold behavior on real, imbalanced crime data.

Index Terms—crime classification, class imbalance, gradient boosting, ROC-AUC, statistical significance, spatio-temporal features

I. INTRODUCTION

Police departments rely on historical incident records, but each record carries one of dozens of fine-grained offense labels, which is too granular for direct operational use. Grouping labels into broader categories is a common first step, yet the resulting classes are imbalanced by an order of magnitude, and models trained on them are rarely evaluated beyond a single accuracy figure.

This paper builds a supervised classifier for this task on roughly 1.06 million real incidents from Chicago’s public data portal (2020–2024). An earlier version of this project used a synthetic multi-city dataset with victim age, gender, and race as predictors. It was abandoned because synthetic labels do not reflect real reporting patterns and because victim demographics raise fairness concerns unrelated to the modeling task. Inputs are therefore restricted to incident context: police district, location type, time, season, day of week, and a domestic flag. Property and Violent incidents together form about 89% of the data, and this imbalance is treated as part of the problem, not hidden behind a single weighting scheme.

Three gaps motivate the design. First, studies combining multiple data sources or spatial structure [1]–[4] usually report one aggregate score, so it is unclear whether gains hold across categories. Second, most classifiers are evaluated at a single decision threshold [5], [6], which mixes ranking quality with the chosen cutoff. Third, several approaches use inputs tied to individuals [1], [7]. The contributions of this work are:

- A reproducible pipeline for grouped crime-type prediction from incident context only, with every preprocessing step and record count reported.
- A separation of ranking quality (ROC-AUC) from fixed-threshold performance (F1, recall), showing that the rarest categories are ranked well despite low default-threshold F1.
- A matched comparison of unweighted and sample-weighted training across five classical models and three neural baselines (MLP, transformer, and graph neural network), with paired significance tests on the model ranking.

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TABLE I
COMPARISON WITH PRIOR CRIME PREDICTION STUDIES

Study	Year	Scope	Imbalance	AUC	Demogr.
Bogomolov et al. [1]	2014	Single city	No	No	Yes
Wang et al. [2]	2016	Single city	No	No	No
Kang & Kang [7]	2017	Single city	No	No	No
Kadar et al. [10]	2019	Multi-region	Yes	No	No
Zhou et al. [11]	2021	Single city	No	No	No
Xia et al. [3]	2021	Multi-city	No	No	No
Qin et al. [13]	2025	Multi-city	N/S	No	No
Shan et al. [14]	2025	N/S	N/S	No	No
Riyazuddin et al. [15]	2026	N/S	N/S	N/S	No
This work	2026	Single city	Yes	Yes	No

N/S = not stated; Demogr. = demographic features used.

II. RELATED WORK

Early work combined non-traditional data with classical models. Bogomolov et al. paired mobile network activity with demographics for London hotspot classification [1]. Wang et al. estimated community-level crime in Chicago from venue records and taxi trip flows [2], and Kang and Kang fused spatial, temporal, and image context in a deep network [7]. Mohler and Porter [8] and Danlin and Fang [9] target area-level rates rather than incident-level categories. Kadar et al. handled class imbalance with an imbalance-aware hyper-ensemble [10]; this paper instead uses sample weighting with a matched unweighted control.

Recent work favors graph and sequence architectures: hypergraph networks for multiplex region relations [3], the homophily-aware HAGEN model [4], Escort for heterogeneous open data [11], meta-path explainable prediction [12], and transformer-graph hybrids [13]–[15]. These mostly report regression-style forecasting errors (MSE/MAE) rather than per-class classification metrics. Reviews by Kounadi et al. [5] and Mandalapu et al. [6] identify inconsistent class-level reporting, class imbalance, and weak cross-study comparability as open problems. Safat et al. and Xu et al. benchmark ML and deep learning methods [16], [17], and Tam and Tanriöver add social-media text [18]. Rotaru et al. show that event-level models can expose enforcement bias [19], and Flaxman et al. show that sparse categories need different handling from frequent ones [20]. Table I compares representative studies with this work.

III. METHODOLOGY

A. Dataset and Preprocessing

All incidents were pulled from the public crime log of the Chicago Data Portal [21], one calendar year at a time through its Socrata interface, covering 2020–2024. Table II lists every preprocessing step with the number of records it keeps. The date string is parsed into a timestamp; records with an unparseable date or a missing date, primary type, district, or location description are dropped. Only the location description has missing values (6,913 records, 0.58%). No

TABLE II
PREPROCESSING PIPELINE AND RECORD COUNTS

Step	Records
Raw download, 2020–2024 (13 columns)	1,185,233
Drop missing date/type/district/location	–6,913
Drop duplicate incident ids	–0
Cleaned incidents	1,178,320
Drop “Other” group (Table III)	–117,519
Modeling dataset	1,060,801
Training set (80%, stratified)	848,640
Test set (20%, stratified)	212,161

TABLE III
TARGET CATEGORIES, GROUPED PRIMARY TYPES, AND CLASS SHARES

Category	Chicago primary types (abridged)	Share
Property	Theft and vehicle theft; burglary; arson; property damage; trespass	51.55%
Violent	Battery and assault; robbery; homicide; sexual offenses; kidnapping and trafficking; intimidation and stalking; offenses against children	37.13%
White-Collar	Deceptive practice; fraud	7.68%
Drug/Public Order	All narcotics offenses; liquor and gambling; prostitution; obscenity and indecency; breaches of the peace; obstructing an officer	3.64%

duplicate incident ids were found. No imputation is applied, so every modeled record is complete.

The target groups Chicago’s 30-plus primary types into four categories, following FBI Part I/II logic (Table III). Types that do not fit a category (other offense, weapons and concealed-carry violations, non-criminal, ritualism) form an “Other” group, which is removed because it has no coherent semantic definition. A stratified four-to-one train/test split (seed 42) is used, and encoders and scalers are fitted on training data alone.

B. Feature Extraction

Eleven features are built from each record (Table IV). Hour and month are cyclical, so each is mapped onto the unit circle so that, for example, 23:00 and 00:00 are adjacent:

$$x_{\sin} = \sin\left(\frac{2\pi t}{T}\right), \quad x_{\cos} = \cos\left(\frac{2\pi t}{T}\right), \quad (1)$$

where t is the hour ($T = 24$) or month ($T = 12$). The four resulting values are standardized with training-set statistics,

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}. \quad (2)$$

The location description has more than 180 distinct values; the 20 most frequent cover 89.6% of incidents and are kept, and the rest are mapped to “Other”. All categorical features are one-hot encoded, giving 66 input columns. Multinomial Naive Bayes requires non-negative inputs, so it receives the one-hot columns only. Victim demographic fields are never used.

TABLE IV
INPUT FEATURES AND ENCODING

Feature	Encoding	Columns
Police district	One-hot	24
Location description (top 20 + Other)	One-hot	21
Day of week	One-hot	7
Season	One-hot	4
Weekend flag (Sat/Sun)	One-hot	2
Night flag (22:00–04:59)	One-hot	2
Domestic flag (from source)	One-hot	2
Hour (sin, cos)	Eq. (1), (2)	2
Month (sin, cos)	Eq. (1), (2)	2
Total		66

C. Models and Class Weighting

All five classical models share the same inputs: Logistic Regression (up to 2000 iterations); one Decision Tree capped at depth 20 with leaves of 50 or more; multinomial Naive Bayes; a 200-tree Random Forest [22] (depth up to 25, leaves of 20 or more); and XGBoost [23] with 300 histogram-based trees (depth 8, step size 0.1, 0.8 row and column sampling). All but XGBoost use scikit-learn [24]. A majority-class classifier (always Property) gives the floor of 51.55% accuracy.

Three neural baselines use the same features. The MLP has two hidden layers (128 and 64 ReLU units). The FT-Transformer [25] embeds each of the eleven features as a 64-dimensional token and passes the token sequence, with a classification token, through three transformer layers with eight attention heads. The District-GCN represents the 24 police districts as graph nodes, each linked to its four nearest districts by the great-circle distance between district centroids (computed from incident coordinates), giving 57 undirected edges (one rarely used district without coordinates keeps only a self-loop). Two graph-convolution layers [26] compute district embeddings,

$$H^{(l+1)} = \sigma(\hat{A}H^{(l)}W^{(l)}), \quad \hat{A} = \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}, \quad (3)$$

in which $\tilde{A}$ is the district link matrix with self-loops added, $\tilde{D}$ holds the row sums of $\tilde{A}$ on its diagonal, each $W^{(l)}$ is trainable, and σ is ReLU (dropped after the final layer). Each incident’s district embedding is concatenated with embeddings of its other features and classified by a two-layer MLP. Adam(W) with step size 10^{-3} is used, and training stops when performance on a held-back tenth of the training data stalls (cap: 20 epochs, 30 for the MLP).

Each model is trained twice: on the raw class distribution (Unweighted) and with balanced per-sample weights (Balanced),

$$w_i = \frac{N}{K N_{y_i}}, \quad (4)$$

for N training incidents, $K = 4$ classes, and N_{y_i} incidents sharing the label of incident i . Sample-level weights are used instead of each model’s class-weight option because Multinomial Naive Bayes has no such option; this keeps the comparison identical across models.

D. Evaluation Metrics and Significance Tests

Test incidents take no part in training. Reported scores are accuracy; balanced accuracy (class-averaged recall); macro (class-uniform) and weighted (size-proportional) precision and F1; the Matthews correlation coefficient (MCC); and class-averaged one-vs-rest ROC-AUC. Macro-F1 averages, over classes, the harmonic mean of precision P_k and recall R_k :

$$F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \left(\frac{P_k^{-1} + R_k^{-1}}{2} \right)^{-1}. \quad (5)$$

Let S be the number of test incidents, C the number classified correctly, and $\mathbf{a}$ and $\mathbf{b}$ the vectors of actual and predicted counts per class. The multiclass MCC is then

$$\text{MCC} = \frac{CS - \mathbf{a}^\top \mathbf{b}}{\sqrt{(S^2 - \|\mathbf{b}\|^2)(S^2 - \|\mathbf{a}\|^2)}}. \quad (6)$$

Three procedures test whether differences between models are real. First, 5-fold stratified cross-validation on a 25% stratified training subsample gives the mean $\bar{x}$ and standard deviation s_x of macro-F1, with the 95% interval

$$\bar{x} \pm t_{4,0.975} \frac{s_x}{\sqrt{5}}, \quad t_{4,0.975} = 2.776. \quad (7)$$

Second, 1000 bootstrap resamples of the test set, shared by all models so that comparisons are paired, give percentile 95% intervals for each metric and for the XGBoost-minus-baseline difference. Third, McNemar’s test compares XGBoost with each other model on the same test incidents:

$$\chi^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}}, \quad (8)$$

where n_{01} (n_{10}) counts incidents that only XGBoost (only the other model) classifies correctly; χ^2 has one degree of freedom under the null hypothesis of equal error rates.

IV. RESULTS

Table V reports test-set performance for the five classical models. XGBoost has the best accuracy, balanced accuracy, macro-F1 (Eq. (5)), weighted-F1, and MCC (Eq. (6)) in the Unweighted configuration, and all models clearly beat the majority baseline. Cross-validated macro-F1 gives the same ranking: XGBoost 0.5044 ± 0.0052 , Decision Tree 0.4939 ± 0.0066 , Random Forest 0.4667 ± 0.0072 , Logistic Regression 0.4383 ± 0.0049 , Naive Bayes 0.4288 ± 0.0018 . By Eq. (7), the XGBoost interval $[0.4980, 0.5109]$ overlaps slightly with the Decision Tree interval $[0.4857, 0.5021]$ but not with any other model.

A. Statistical Significance

Because cross-validation on five folds cannot separate XGBoost from the Decision Tree, the pipeline was re-executed end to end to obtain test-set predictions for paired tests, with the three neural baselines added. Random Forest was omitted because of its training cost (about 26 minutes versus 3 minutes for XGBoost). The re-run used a later snapshot of the same portal query (1,062,495 modeled incidents, test $n = 212,499$),

TABLE V
TEST-SET PERFORMANCE UNDER THE UNWEIGHTED AND BALANCED (SAMPLE-WEIGHTED) CONFIGURATIONS ($n = 212,161$)

Model	Unweighted						Balanced					
	Acc	BA	P _M	F1 _M	F1 _W	MCC	Acc	BA	P _M	F1 _M	F1 _W	MCC
Logistic Regression	0.6620	0.4094	0.6227	0.4339	0.6322	0.3902	0.4876	0.5851	0.4616	0.4224	0.5373	0.3267
Decision Tree	0.6711	0.4672	0.6198	0.5045	0.6544	0.4147	0.5251	0.6010	0.4662	0.4515	0.5661	0.3469
Naive Bayes	0.6478	0.4058	0.5884	0.4262	0.6170	0.3609	0.4872	0.5684	0.4526	0.4174	0.5353	0.3156
Random Forest	0.6721	0.4453	0.6489	0.4829	0.6488	0.4136	0.5417	0.6101	0.4746	0.4632	0.5799	0.3591
XGBoost	0.6747	0.4687	0.6341	0.5078	0.6576	0.4205	0.5366	0.6140	0.4719	0.4616	0.5752	0.3590
Majority baseline	0.5155	0.2500	0.1289	0.1701	0.3507	0.0000	–	–	–	–	–	–

Acc = accuracy; BA = balanced accuracy; P = precision; subscripts M and W = macro and weighted averages.
Macro recall equals BA and is omitted.

TABLE VI
PAIRED SIGNIFICANCE TESTS AGAINST XGBOOST (RE-RUN, TEST $n = 212,499$)

Model	F1 _M [95% CI]	χ^2	p	$\Delta F1_M$ [95% CI]
<i>Unweighted</i>				
XGBoost	0.505 [0.502, 0.509]	–	–	–
MLP	0.504 [0.500, 0.508]	0.5	0.468	[0.000, 0.004]
FT-Transformer	0.494 [0.491, 0.498]	9.9	0.002	[0.009, 0.013]
District-GCN	0.493 [0.489, 0.496]	7.4	0.006	[0.011, 0.015]
Decision Tree	0.499 [0.495, 0.503]	51.9	<0.001	[0.005, 0.009]
Log. Reg.	0.436 [0.432, 0.439]	317.3	<0.001	[0.067, 0.073]
Naive Bayes	0.428 [0.425, 0.432]	1046.9	<0.001	[0.074, 0.080]
<i>Balanced</i>				
XGBoost	0.461 [0.458, 0.463]	–	–	–
MLP	0.450 [0.448, 0.452]	624.6	<0.001	[0.009, 0.012]
FT-Transformer	0.441 [0.439, 0.443]	2216.2	<0.001	[0.019, 0.021]
District-GCN	0.452 [0.450, 0.455]	259.2	<0.001	[0.007, 0.009]
Decision Tree	0.454 [0.451, 0.456]	89.6	<0.001	[0.006, 0.008]
Log. Reg.	0.422 [0.419, 0.424]	3617.9	<0.001	[0.038, 0.040]
Naive Bayes	0.418 [0.416, 0.420]	2777.7	<0.001	[0.041, 0.044]

χ^2 , p : McNemar test, Eq. (8). $\Delta F1$: XGBoost minus model, paired bootstrap (1000 resamples).

because the portal revises records over time. Re-run metrics stay within 0.006 of Table V for every model and preserve the model ranking. Table VI shows the results.

Against the three weaker classical models (Naive Bayes, Logistic Regression, and the Decision Tree), XGBoost wins in both configurations ($p < 0.001$), and each of these bootstrap difference intervals lies entirely above zero. The gap to the Decision Tree is small (0.5–0.9 macro-F1 points), but on more than 200,000 paired test incidents it is consistent. The test-set bootstrap intervals for accuracy are narrow (about ± 0.002 for every model), so accuracy differences above roughly 0.004 are reliable. The MLP is the only model that XGBoost does not beat in the Unweighted configuration: their accuracies (0.6720 versus 0.6717) do not differ under McNemar’s test ($p = 0.47$), and the macro-F1 advantage of XGBoost is marginal (interval [0.000, 0.004]). Under balancing, XGBoost is again significantly better than the MLP. The FT-Transformer and District-GCN reach the same accuracy as each other (0.6708) but lower macro-F1 than XGBoost, and XGBoost is significantly better than both in both configurations ($p \leq 0.006$). Sharing

information between neighboring districts through Eq. (3) therefore adds nothing beyond a separate representation per district: the per-district signal is already strong (Section IV-E), and 24 districts are few enough to estimate individually from about 850,000 training incidents. On these tabular, mostly categorical features, none of the three neural models outperforms gradient boosting, while they give less direct feature attributions and, for the transformer, need about 18 times longer training (599 s versus 33 s).

B. Model Comparison

XGBoost fits sequential corrections to residual error, so later trees learn interactions such as which location types dominate in which districts. Logistic Regression and Naive Bayes cannot represent the domestic–location interaction that dominates feature importance (Section IV-E), and Naive Bayes is weakest on every metric because its conditional-independence assumption is violated: location values concentrate in particular districts, and the domestic flag correlates with several location categories. The single Decision Tree comes within 0.4 accuracy points of XGBoost because most signal lies in a few high-value categorical splits (domestic, district 11), which a depth-20 tree already captures. Random Forest averages its trees and so does not keep targeting minority-class errors as boosting does, and it takes nearly nine times longer to train than XGBoost.

C. Impact of Class Imbalance

Balancing with the weights of Eq. (4) lowers accuracy for all models, from 0.65–0.67 to 0.49–0.54, and raises balanced accuracy from 0.41–0.47 to 0.57–0.61. Macro-F1 does not improve; for XGBoost it falls from 0.5078 to 0.4616. The cause is visible at class level. For XGBoost, Drug/Public Order recall rises from 0.19 to 0.67 while its precision falls from 0.61 to 0.15; White-Collar recall rises from 0.27 to 0.75 while precision falls from 0.54 to 0.27. Balancing therefore moves the operating point toward recall at the cost of many false positives, rather than producing a better classifier. The preferred point depends on whether the application tolerates false alarms to catch more minority-class incidents.

D. Per-Class Performance and Ranking Quality

Table VII reports per-class results for XGBoost. Property and Violent are predicted well at the default threshold (F1

TABLE VII
XGBOOST PER-CLASS RESULTS (ORIGINAL RUN)

Category	Unweighted			Balanced	
	P	R	F1	R	AUC
Property	0.66	0.84	0.74	0.50	0.770
Violent	0.73	0.57	0.64	0.54	0.794
White-Collar	0.54	0.27	0.36	0.75	0.876
Drug/Public Order	0.61	0.19	0.29	0.67	0.859

P = precision; R = recall; AUC: one-vs-rest, Unweighted.

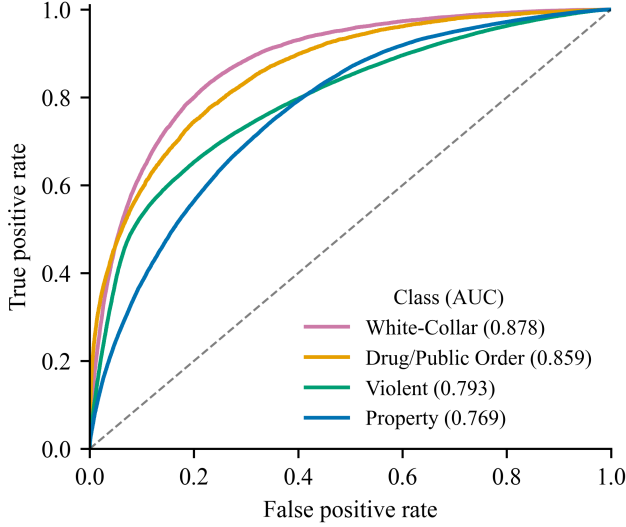


Fig. 1. One-vs-rest ROC curves for XGBoost, Unweighted configuration (re-run snapshot). Legend values are per-class AUC.

0.74 and 0.64), whereas Drug/Public Order and White-Collar are not (0.29 and 0.36). The AUC order is the reverse (Fig. 1): White-Collar (0.876) and Drug/Public Order (0.859) are ranked better than Violent (0.794) and Property (0.770). The model separates the rare classes from the rest better than it separates the two common classes from each other, but their low base rates (7.68% and 3.64%) push the argmax decision rule toward under-predicting them. The weakness is a thresholding and base-rate effect, not a lack of signal. The re-run reproduces these AUCs within 0.002.

E. Feature Importance

Fig. 2 shows the 15 features with the highest XGBoost gain importance (Balanced configuration). The domestic flag dominates, followed by district 11 and several location categories (residence, sidewalk, apartment, street). None of the four cyclical time features appear in the top 15. This is not because incidents are spread evenly in time: Fig. 3 shows volume rising from 08:00, staying high from about 15:00 to 20:00, and reaching its minimum near 05:00 on every day. The sharp columns at 00:00 and 12:00 are recording artifacts: 3.8% and 2.6% of all records carry a timestamp of exactly 00:00 or 12:00, which suggests a default value entered when

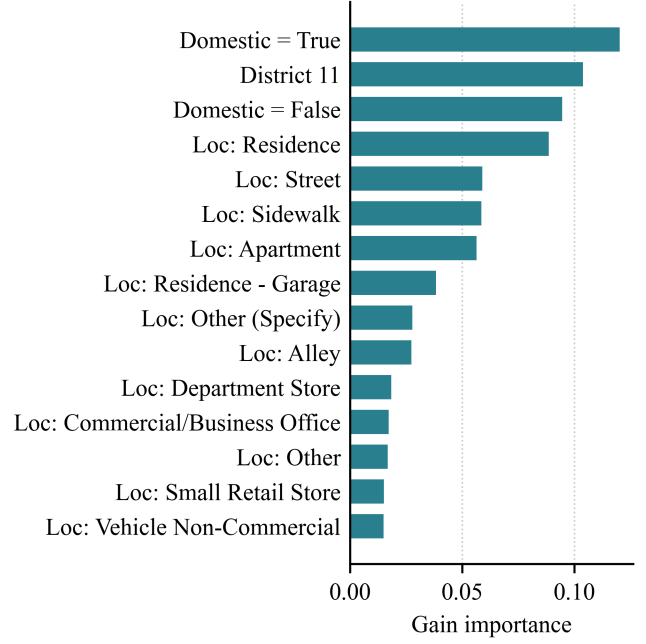


Fig. 2. Top-15 features by XGBoost gain importance, Balanced configuration (re-run snapshot; same 15 features as the original run).

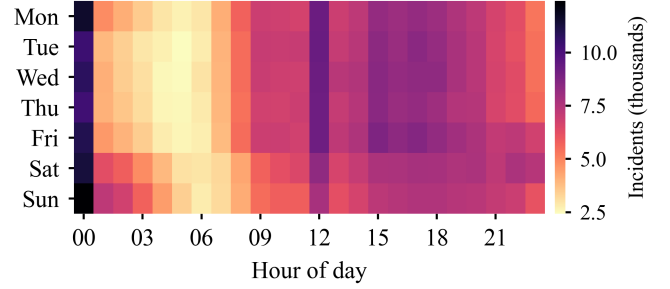


Fig. 3. Incident volume by hour of day and day of week, all categories pooled (re-run snapshot). Darker cells indicate more incidents.

the true time is unknown (deceptive practice is the most common type at 00:00). The temporal pattern is similar across crime categories, however, so it does little to separate them. The learned signal is primarily spatial and contextual, and this study does not support a claim of strong time-of-day or seasonal predictive power for category classification.

V. LIMITATIONS AND FUTURE WORK

Single city. Chicago's district boundaries and reporting conventions shape the features directly, as the role of district 11 shows. Validating the pipeline on open data from other cities, following the multi-city approach of Rotaru et al. [19], or fine-tuning a Chicago model on a smaller second-city dataset [27], would test whether the findings generalize.

Approximate spatial adjacency. The District-GCN links districts by centroid proximity, which approximates but does

not equal boundary contiguity, and it uses a fixed graph. Exact boundary adjacency, finer spatial units such as beats or community areas, and learned or time-varying graphs [3], [4] may capture neighborhood effects that 24 coarse districts cannot.

Per-incident deep models only. The neural baselines classify single incidents. Spatio-temporal graph-transformer models [13], [14] operate on sequences of area-level counts and were not benchmarked; they become comparable once the forecasting task below is defined.

No forecasting. Each incident is classified independently, and 2020–2024 is pooled, including COVID-era anomalies, without modeling trends. The results are not forecasts. Predicting district-level counts per category with recurrent or temporal transformer models would extend the pipeline from classification to forecasting.

Thresholds and attribution. Because the rare classes are already ranked well, per-class thresholds or calibrated probabilities (Platt scaling, isotonic regression) are likely to outperform sample weighting, and SHAP attributions would give a finer diagnostic than aggregate gain importance. The placeholder timestamps at 00:00 and 12:00 were kept in this study and may weaken the time features; flagging or excluding them is a simple refinement.

VI. CONCLUSION

This paper classified 1,060,801 Chicago crime incidents (2020–2024) into Violent, Property, Drug/Public Order, and White-Collar categories using eleven incident-context features and no victim demographics. XGBoost was the best classical model in the unweighted configuration (67.47% accuracy, macro-F1 0.5078, cross-validated macro-F1 0.5044 ± 0.0052), and paired McNemar and bootstrap tests showed its margin over Naïve Bayes, Logistic Regression, and the Decision Tree to be significant ($p < 0.001$). Three neural baselines (an MLP, an FT-Transformer, and a district-adjacency graph convolutional network) matched or trailed XGBoost; none exceeded it. The two rarest categories had the highest AUCs (0.876 and 0.859), so their low F1 reflects base rates and thresholds rather than missing signal. Sample weighting traded precision and accuracy for minority-class recall without improving macro-F1. The learned signal was mainly spatial and contextual, led by the domestic flag, police district, and location type.

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